<table>
<tr><th></th><th></th><th></th></tr>
<tr><td rowspan="2">2(a)(i)</td><td>1. energy transfer to the system by heating</td><td>B1</td></tr>
<tr><td>2. (external) work done on the system</td><td>B1</td></tr>
<tr><td>2(a)(ii)</td><td>decrease in internal energy</td><td>B1</td></tr>
<tr><td rowspan="2">2(b)(i)</td><td>no change (in internal energy)</td><td>B1</td></tr>
<tr><td>(because) no change in temperature</td><td>B1</td></tr>
<tr><td rowspan="2">2(b)(ii)</td><td>work done $= p\Delta V$

$= (–)1.6 \times 10^{5} \times (2.4 – 0.87) \times 10^{–3}$</td><td>C1</td></tr>
<tr><td>$= (–)240$ J</td><td>A1</td></tr>
<tr><td rowspan="4">2(b)(iii)</td><td>first row all correct (0, 480, 480)</td><td>A1</td></tr>
<tr><td>second row all correct (–1100, 0, –1100)</td><td>A1</td></tr>
<tr><td>final column of third row calculated correctly from the two values above it, so that the final column adds up to 0</td><td>A1</td></tr>
<tr><td>second column in final row correct, with correct negative sign
and
first column in final row calculated correctly so that it adds to the second column to give the third column

(fully correct table is:
<table><tr><td>0</td><td>480</td><td>480</td></tr><tr><td>–1100</td><td>0</td><td>–1100</td></tr><tr><td>860</td><td>–240</td><td>620</td></tr></table>)</td><td>A1</td></tr>
</table>